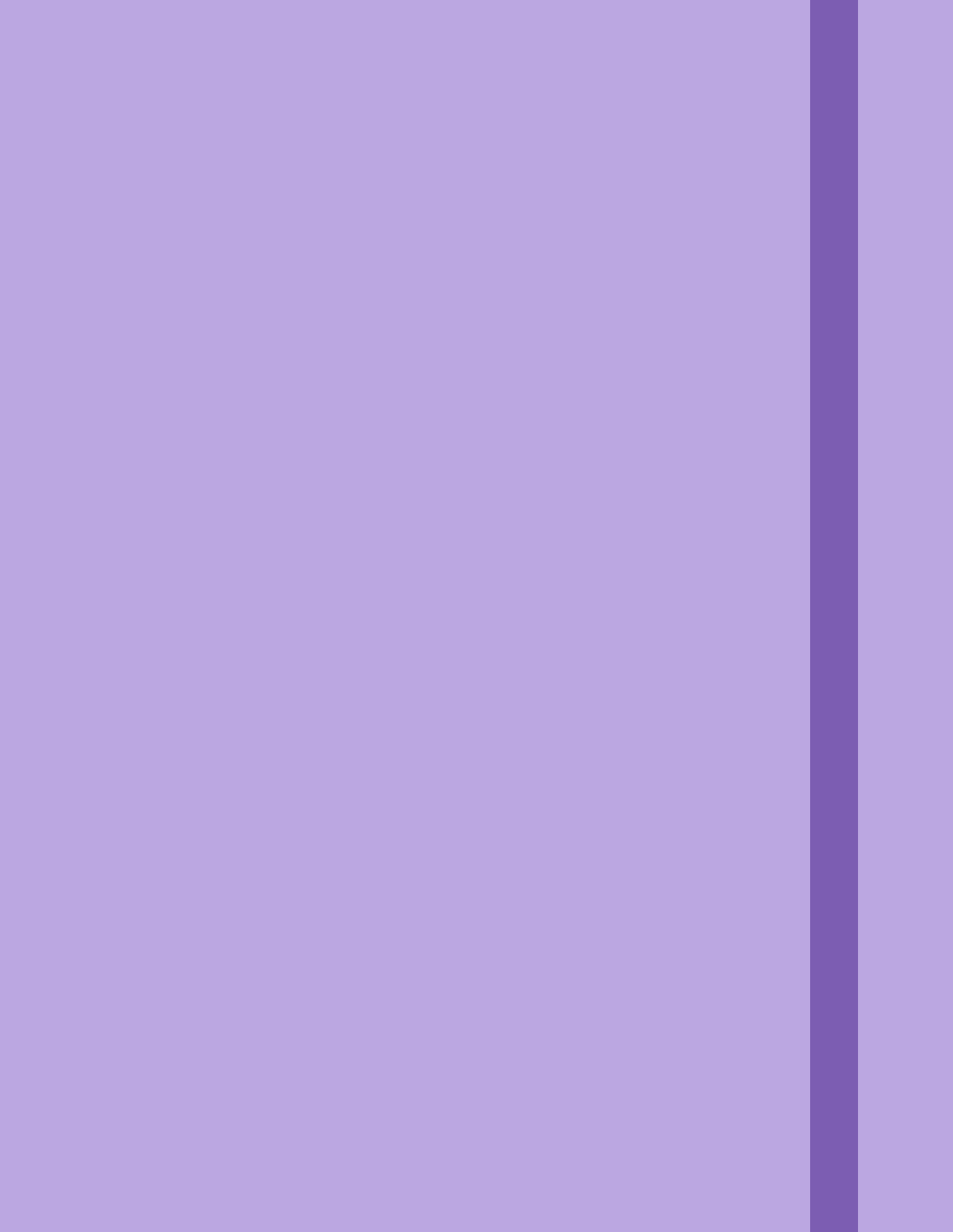




Probability

Algebraic Variables $\rightarrow$ In Algebra variable, like x , is an unknown value.

Random Variable $\rightarrow$ A set of possible value from a random experiment.

eg) Tossing a coin

$$H=1, T=0$$

$$x = \{1, 0\}$$

eg.) Rolling a dice.

$$\gamma = \{1, 2, 3, 4, 5, 6\}$$

randomly.

sample space

- Denoted by capital letters.

Types of Random Variable :-

- Discrete Random Variable.

g) $\{H, T\}$, $\{1, 2, 3, 4\}$

- Continuous Random Variable.

g) $\{ 8.32, 8.50, \dots \}$
Range: 8-9

Probability Distributions $\rightarrow$ List of all possible outcomes of random variables along with their corresponding probability values.

ex.)	Coin Toss	1 (H)	0 (T)
	Probability	$\frac{1}{2}$	$\frac{1}{2}$

[illegible]

eg) Rolling 2 dice

{ 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 }

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$$P(2) = \frac{1}{36}$$

$$P(3) = \frac{2}{36}$$

$$P(4) = \frac{3}{36}$$

$$P(5) = \frac{4}{36}$$

$$P(6) = \frac{5}{36}$$

$$P(7) = \frac{6}{36}$$

$$P(8) = \frac{5}{36}$$

$$P(9) = \frac{4}{36}$$

$$P(10) = \frac{3}{36}$$

$$P(11) = \frac{2}{36}$$

$$P(12) = \frac{1}{36}$$

Problem with Distribution $\rightarrow$ In many case, no. of outcomes are very large.

eg) Rolling 10 dice.

It could be ∞ .

Solution $\rightarrow$ Function ?

Can we use fn. to model relationship b/w outcome and probability?

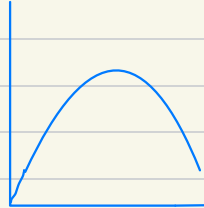
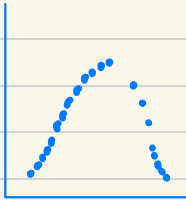
$$y = f(x)$$

* Plotting $\uparrow$ the graph of fn. we can visualize the shape of probability distribution.
Probability Distribution Function. $\uparrow$

Types of PDF (Probability Distribution Function).

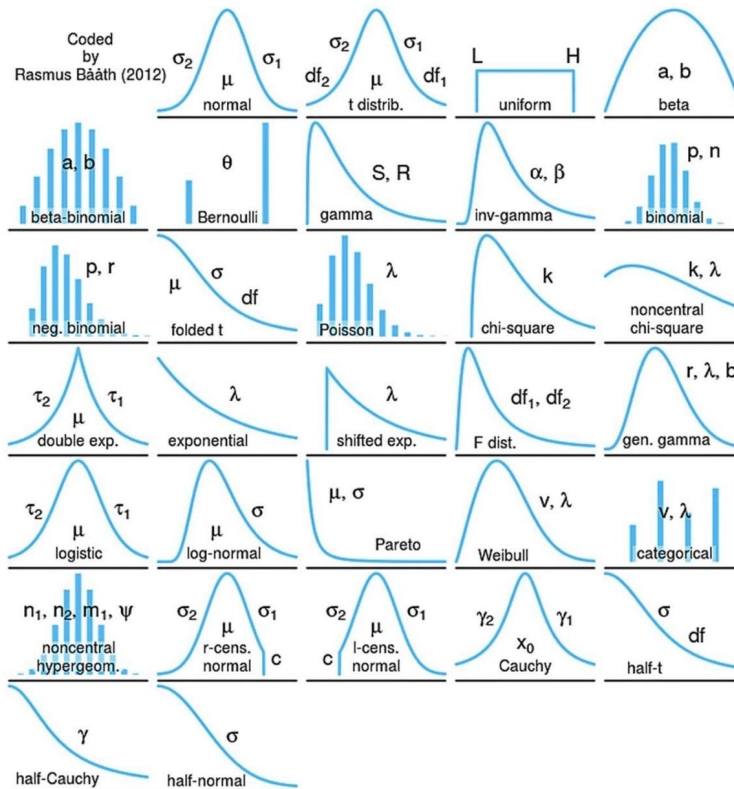
Discrete

Continuous.



Probability Distributions

Coded by
Rasmus Bááth (2012)



Why a Probability Distribution important?

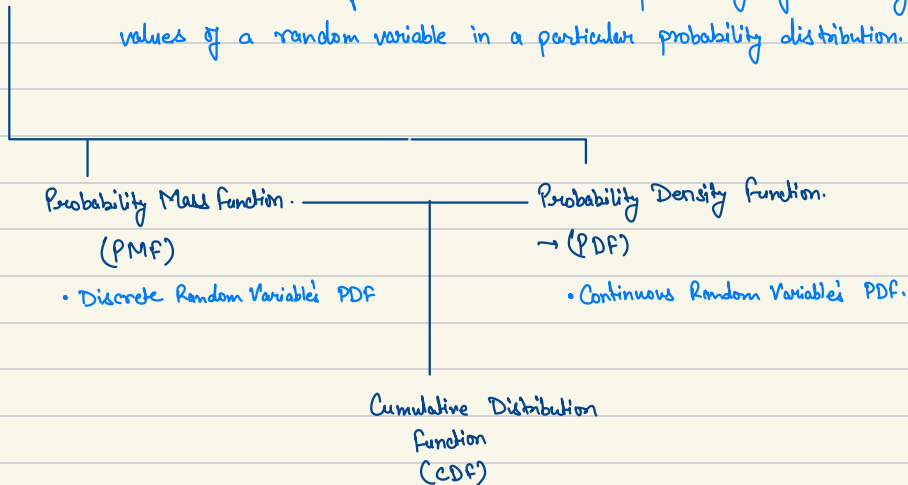
- Gives an idea about the shape/distribution of the data.
- And if our data follows a famous distribution then we automatically know about the data.

Parameters of PDF.

Parameters in PDF are numerical value that determines the shape, location and the scale of the distribution.

Different PDF have different set of parameters that determine their shape and characteristics, and understanding these parameters is essential in statistical analysis & inference.

PDF $\rightarrow$ is a mathematical fn that describes the probability of obtaining different values of a random variable in a particular probability distribution.



Now, we use PDF as Probability Density Function.

Probability Mass Function (PMF)

It's a mathematical fun. that describes the probability distribution of a discrete random variable.

The PMF of a discrete random variable assigns a probability to each value of the random variable. The probabilities assigned by PMF must satisfy two conditions:

- Probability assigned to each value must be non-negative.
- The sum of probability assigned to all possible values must equal 1.

eg)

1	2	3	4	5	6
$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$y = \begin{cases} \frac{1}{6}, & \text{if } x \in \{1, 2, 3, 4, 5, 6\} \\ 0, & \text{otherwise} \end{cases}$$

eg)

- Bernoulli Distribution.
- Binomial Distribution.

Cumulative Density Function (CDF):

CDF of PMF :-

The CDF $F(x)$ describes the probability that a random variable X with a given probability distribution will be found at a value less than or equal to x .

$$F(x) = P(X \leq x)$$

$$\begin{aligned} \text{eg)} \quad F(X \leq 4) &= f(x=4) + f(x=3) + f(x=2) + f(x=1) \\ &= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} \\ &= \frac{4}{6} = \frac{2}{3}. \end{aligned}$$

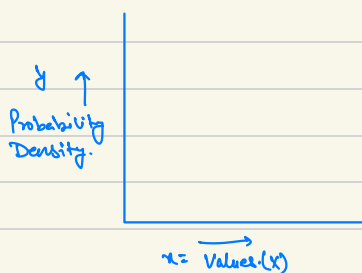
eg)

	1	2	3	4	5	6
PMF	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
CDF	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	$\frac{6}{6}$

0 1

Probability Density Function. (PDF)

It describes the probability distribution of continuous random variable.



eg) Normal Distribution.

log-Normal Distribution.

Poisson Distribution.

Density Estimation.

Density estimation is a statistical technique used to estimate the probability density function (PDF) of a random variable based on a set of observations or data. In simpler terms, it involves estimating the underlying distribution of a set of data points.

Density estimation can be used for a variety of purposes, such as **hypothesis testing**, **data analysis**, and **data visualization**. It is particularly useful in areas such as **machine learning**, where it is often used to estimate the probability distribution of input data or to model the likelihood of certain events or outcomes.

There are various methods for density estimation, including **parametric** and **non-parametric approaches**. Parametric methods assume that the data follows a specific probability distribution (such as a normal distribution), while non-parametric methods do not make any assumptions about the distribution and instead estimate it directly from the data.

Commonly used techniques for density estimation include **kernel density estimation (KDE)**, **histogram estimation**, and **Gaussian mixture models (GMMs)**. The choice of method depends on the specific characteristics of the data and the intended use of the density estimate.

Parametric
Non Parametric
Density Estimation

Parametric Density Estimation.

It's a method of estimating the probability density function (PDF) of a random variable by assuming that the underlying distribution belongs to a specific parametric family of probability distributions, such as the normal, exponential or Poisson distributions.

Non-Parametric Density Estimation.

KDE $\rightarrow$ (Kernel Density Estimation.)

But sometimes the distribution is not clear or it's not one of the famous distributions.

Non-parametric density estimation is a statistical technique used to estimate the probability density function of a random variable without making any assumptions about the underlying distribution. It is also referred to as non-parametric density estimation because it does not require the use of a predefined probability distribution function, as opposed to parametric methods such as the Gaussian distribution.

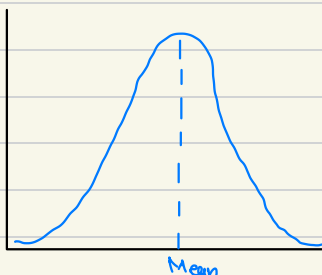
The non-parametric density estimation technique involves constructing an estimate of the probability density function using the available data. This is typically done by creating a **kernel density estimate**

Non-parametric density estimation has several advantages over parametric density estimation. One of the main **advantages** is that it **does not require the assumption of a specific distribution**, which allows for more flexible and accurate estimation in situations where the underlying distribution is unknown or complex. However, non-parametric density estimation can be **computationally intensive and may require more data to achieve accurate estimates** compared to parametric methods.

(Nbk: Watch video for KDE.)

Normal Distribution.

Also known as Gaussian distribution, It's a continuous probability distribution that's symmetrical around mean, with a bell curve.



- Tail
- Asymptotic in nature.
- Lots of points near the mean & very few far away.

Normal distribution is characterized by 2 parameters:

- Mean (μ) $\rightarrow$ Center of distribution.
- Std. (σ) $\rightarrow$ spread of distribution.

Why is it so important?

Common in Nature. Many natural phenomena follow normal distribution. e.g., height of people, weight of objects, etc.

Convenient way to model & analyse such data.

PDF Equation.

$$y = f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}$$

μ = mean σ = Standard Deviation.

grapher $\rightarrow$ desmos.com

$$y = e^x$$

$$y = e^{-x}$$

$$y = e^{-2x}$$

$$y = e^{-(x-\mu)^2}$$

$$y = e^{-\frac{1}{2}(x-\mu)^2}$$

$$y = e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Area under curve ≈ 1

$$y = \int e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Standard Normal Variate / Distribution (Z)

- Special case of normal distribution.

$$\text{if } \mu = 0 \\ \sigma = 1$$

$$\text{then, } \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- It allows us to compare different distributions with each other.
- Calculate probabilities using standardized tables or software.

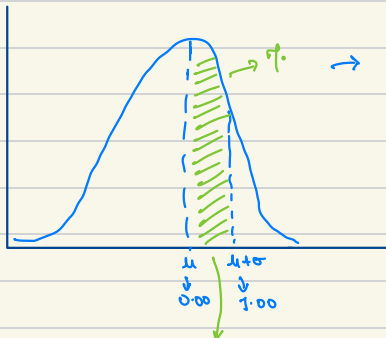
How to transform a normal distribution to Standard Normal Variate.

$$X \sim (\mu, \sigma)$$

$$\frac{x - \mu}{\sigma}$$

Q1) Normal Distribution $X \sim (\mu, \text{std})$

What % of population lies b/w mean and 1 std, 2 std & 3 std?



$$Z = \frac{x - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = 0$$

$$Z = \frac{\mu + \sigma - \mu}{\sigma} = 1$$

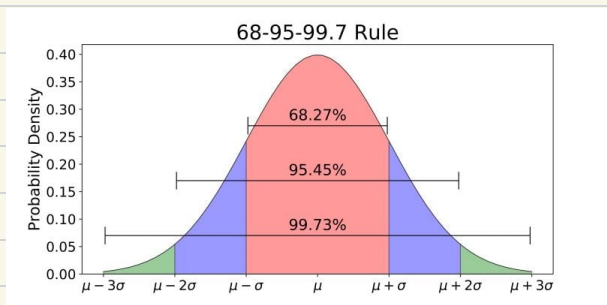
(94-50)% from Z table.

34%

σ to $\sigma \rightarrow 34\%$

Empirical Rule of Subnormal Variat.

→ 68-95-99.7 % Rule.



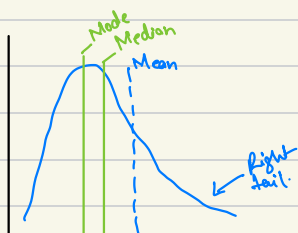
Properties :-

- Symmetric.
- Measure of Central Tendencies are equal i.e. Mean = Median = Mode.
- Empirical Rule.
- Area under the curve = 1 (Generally for all PDF)

Skewness

→ A normal distribution is bell-shaped.

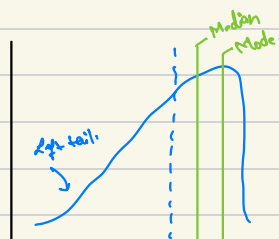
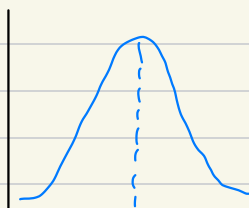
Skewness represents that data is not symmetrical i.e. it's not normally distributed.



+ve Skewed

Right Skewed

mode < median < mean



-ve Skewed

Left Skewed.

mean < median < mode.

- Greater the skewed, the greater the distance b/w mode, median and mean.

How Skewness is calculated?

Statistical moment formula: (Sample Skew) $\rightarrow$ 3rd moment.

$$\frac{n}{(n-1)(n-2)} \sum \left(\frac{(x - \bar{x})}{s} \right)^3$$

$s \rightarrow$ sample std. $n \rightarrow$ no. of points in sample.
 $\bar{x} \rightarrow$ mean

1st moment $\rightarrow$ mean

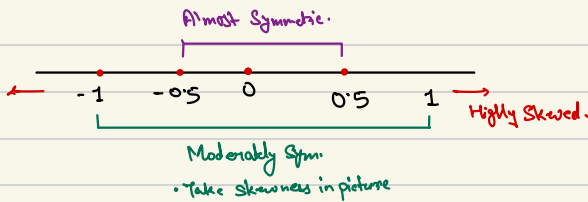
2nd moment $\rightarrow$ variance

3rd moment $\rightarrow$ skew

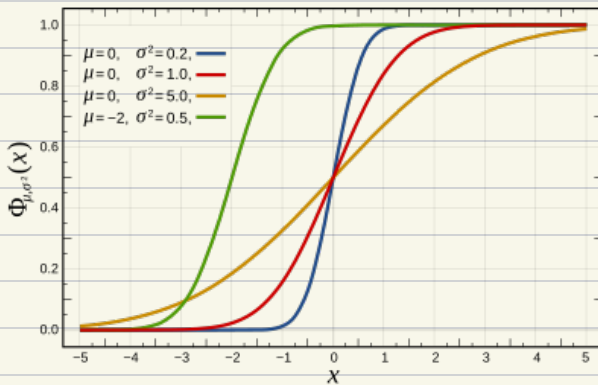
4th moment $\rightarrow$ kurtosis.

- Pearson's skewness $\rightarrow$ not so useful.

★



CDF of Normal Distribution.



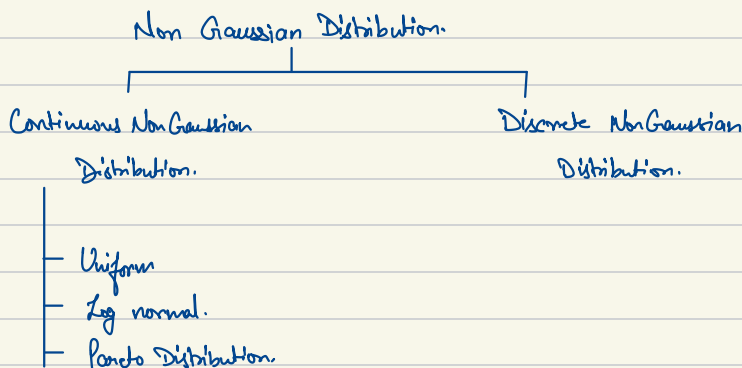
$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

Uses in Data Science :-

- Outliers Detection.
- Assumption on data for ML algo $\rightarrow$ Linear Regression & GMM.
- Hypothesis Testing.
- Central Limit Theorem.

Non Gaussian Distribution.



Given a distribution, how can we tell it's a normal distribution?

Soln $\rightarrow$ Q-Q plot.

Kurtosis.

Kurtosis is the 4th statistical moment. In probability theory & stats, Kurtosis (meaning "curved") is a measure of the "tailedness" of the probability distribution of a real-valued random variable. Like skewness, kurtosis describes a particular aspect of a probability distribution.

Formula.

$$\sum \frac{n * (n+1)}{(n-1) * (n-2) * (n-3)} * \sum \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3 * (n-1)^4}{(n-2) * (n-3)}$$

Practical Use :-

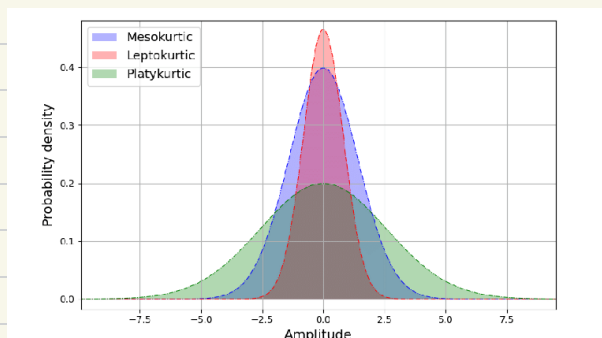
- In finance, Kurtosis risk refers to the risk with the possibility of extreme outcome or "fat tails" in the distribution of returns of a particular asset or portfolio.

If a distribution has high Kurtosis, it means that there is higher likelihood of extreme events occurring, either positive or negative, compared to a normal distribution.

Excess Kurtosis

It's a measure of how much more peaked or flat a distribution is compared to a normal distribution, which is considered to have kurtosis of 0. It's calculated by subtracting 3 from the sample kurtosis coefficient.

Types of Kurtosis :-



Leptokurtic

A distribution with positive excess kurtosis is called leptokurtic. "Lepto-" means "slender". In terms of shape, a leptokurtic distribution has fatter tails. This indicates that there are more extreme values or outliers in the distribution.

Example - Assets with positive excess kurtosis are riskier and more volatile than those with a normal distribution, and they may experience sudden price movements that can result in significant gains or losses.

Sample - 3 > 0, then leptokurtic

Platykurtic

A distribution with negative excess kurtosis is called platykurtic. "Platy-" means "broad". In terms of shape, a platykurtic distribution has thinner tails. This indicates that there are fewer extreme values or outliers in the distribution.

Assets with negative excess kurtosis are less risky and less volatile than those with a normal distribution, and they may experience more gradual price movements that are less likely to result in large gains or losses.

Sample $-3 < 0$, then Platykurtic.

Mesokurtic

Distributions with zero excess kurtosis are called mesokurtic. The most prominent example of a mesokurtic distribution is the normal distribution family, regardless of the values of its parameters.

Mesokurtic is a term used to describe a distribution with a excess kurtosis of 0, indicating that it has the same degree of "peakedness" or "flatness" as a normal distribution.

Example -

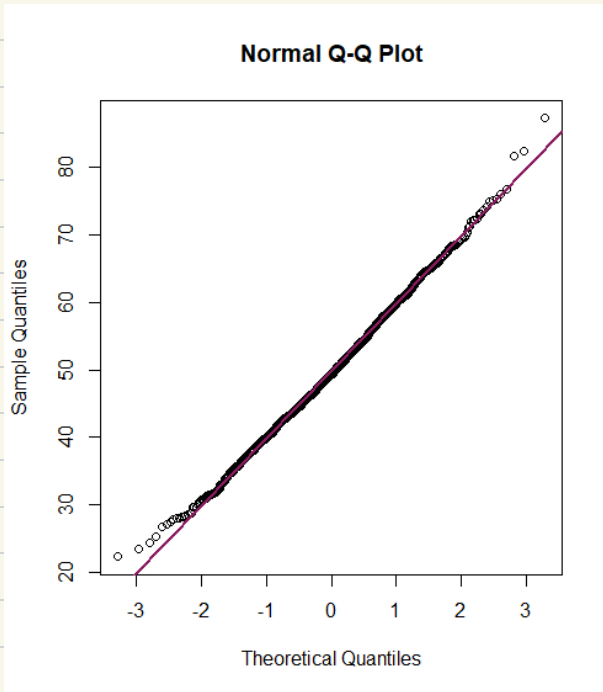
In finance, a mesokurtic distribution is considered to be the ideal distribution for assets or portfolios, as it represents a balance between risk and return.

Q1) How to find if a given distribution is normal or not?

- **Visual inspection:** One of the easiest ways to check for normality is to visually inspect a histogram or a density plot of the data. A normal distribution has a bell-shaped curve, which means that the majority of the data falls in the middle, and the tails taper off symmetrically. If the distribution looks approximately bell-shaped, it is likely to be normal.
- **QQ Plot:** Another way to check for normality is to create a normal probability plot (also known as a Q-Q plot) of the data. A normal probability plot plots the observed data against the expected values of a normal distribution. If the data points fall along a straight line, the distribution is likely to be normal.
- **Statistical tests:** There are several statistical tests that can be used to test for normality, such as the Shapiro-Wilk test, the Anderson-Darling test, and the Kolmogorov-Smirnov test. These tests compare the observed data to the expected values of a normal distribution and provide a p-value that indicates whether the data is likely to be normal or not. A p-value less than the significance level (usually 0.05) suggests that the data is not normal.

Q-Q plot

A Q-Q plot (quantile quantile plot) is a graphical tool used to assess the similarity of the distribution of two set of data. It's particularly useful for determining whether a set of data follows a normal distribution.



Q-Q plot:

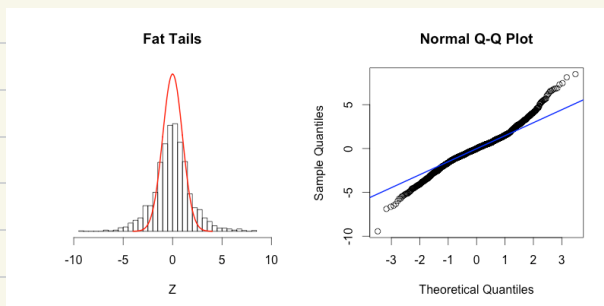
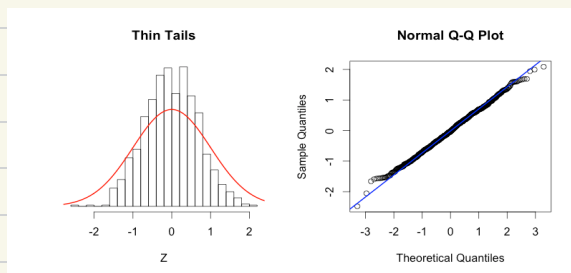
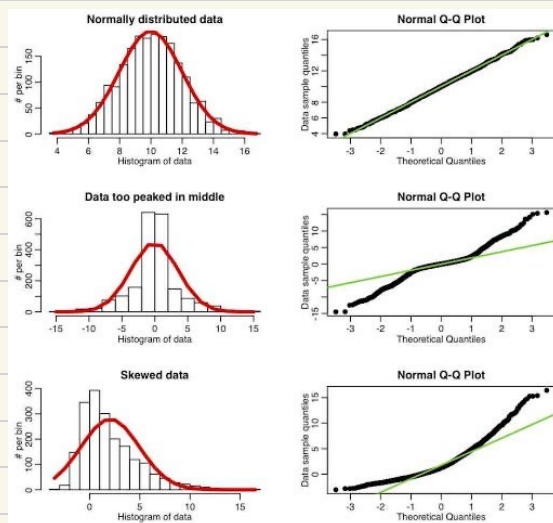
$[X, Y] \rightarrow$ normal distributed
data $\rightarrow$ Yes/No.
 $\downarrow$
 theoretical distribution.

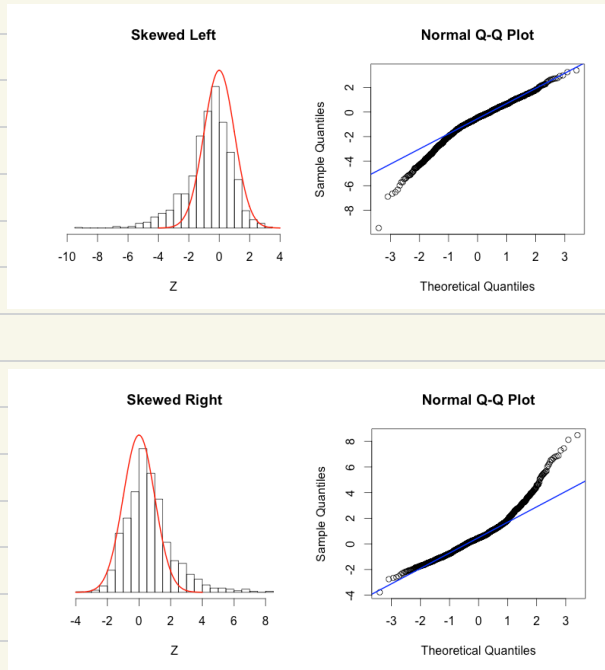
- Theoretical data. (Y)
- $X \rightarrow$ sort (X)
Quantiles $\rightarrow 100$
 $\downarrow$
• same

$\left\{ \begin{array}{l} \bullet Y - \text{quantile} \\ \bullet X - \text{quantile} \end{array} \right\}$
 $\rightarrow$ plot on graph

$\rightarrow$ If all points are in single line.
then $X \& Y$ are normally distributed.

Q-Q plot Interpretation :-



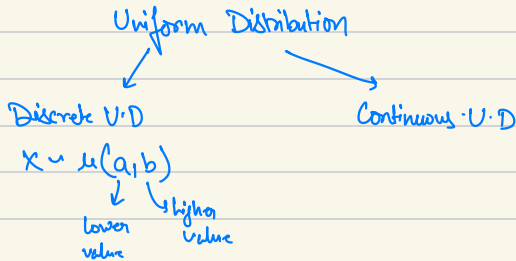


Q1 Does Q-Q plots only detect normal distribution?

$[x, y]$
 ↑
 { Normal
 Uniform
 Pareto
 Log-normal
 etc.

Uniform Distribution.

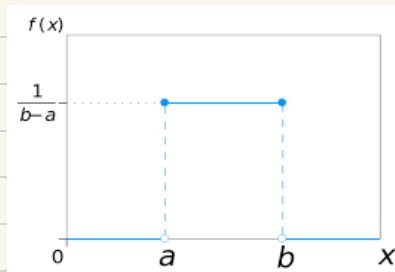
In probability theory & statistics, a uniform distribution is a probability distribution where all outcomes are equally likely or within given range. This means that if you select a random value from this range, any value would be as likely as any other value.



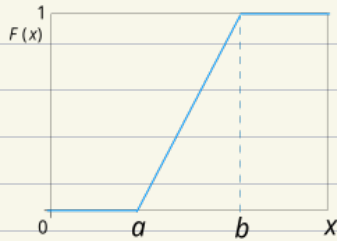
eg.) • Height of a person randomly selected from a group of individuals whose height ranges from 5'6" to 6'0" would follow continuous u.d.

PDF, CDF & Graph.

PDF $\rightarrow$



Cdf $\rightarrow$



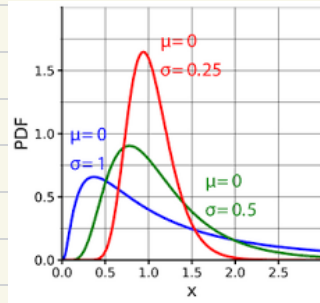
Skewness $\rightarrow 0 \rightarrow$ symmetric in nature.

- Application in Machine learning and Data Science
 - Random initialization:** In many machine learning algorithms, such as neural networks and k-means clustering, the initial values of the parameters can have a significant impact on the final result. Uniform distribution is often used to randomly initialize the parameters, as it ensures that all values in the range have an equal probability of being selected.
 - Sampling:** Uniform distribution can also be used for sampling. For example, if you have a dataset with an equal number of samples from each class, you can use uniform distribution to randomly select a subset of the data that is representative of all the classes.
 - Data augmentation:** In some cases, you may want to artificially increase the size of your dataset by generating new examples that are similar to the original data. Uniform distribution can be used to generate new data points that are within a specified range of the original data.
 - Hyperparameter tuning:** Uniform distribution can also be used in hyperparameter tuning, where you need to search for the best combination of hyperparameters for a machine learning model. By defining a uniform prior distribution for each hyperparameter, you can sample from the distribution to explore the hyperparameter space.

Log - Normal Distribution.

In probability theory & statistics, a log normal distribution is a heavy tailed continuous probability distribution of a random variable whose log is normally distributed.

→ Right Skewed; $\log(X) \sim N(\mu, \sigma)$



eg.) • The length of comments posted in internet discussion forums follows a log-normal distribution.

- Length of chess game.
- In economics, there is evidence that the income of 97% - 99% of the population is distributed log-normally.

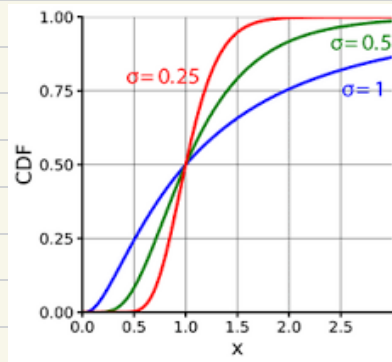
Denoted as

$$X \sim \text{lognormal}(\mu, \sigma)$$

PDF equation.

$$\frac{1}{x\sigma\sqrt{2\pi}} e^{\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right)}$$

Cdf $\rightarrow$



Skewness $\rightarrow$ skewed.

How to check if a random variable is log-normally distributed?

$X \rightarrow$ log normal?

$$\underline{\log(X)} \sim N(\mu, \sigma)$$

$\downarrow$

QQ plot.

$\downarrow$

y -normal then X is log normal.

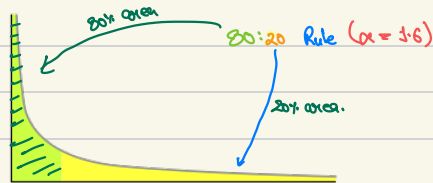
Pareto Distribution.

The Pareto Distribution is a type of probability distribution that is commonly used to model the distribution of wealth, income & other quantities that exhibit a similar power-law behaviour.

What is Power law?

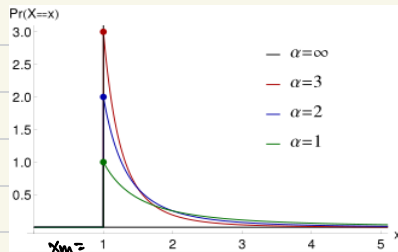
In maths, a power law is a functional relationship b/w 2 variables, where one variable is proportional to a power of the other. Specifically, if y & x are 2 variables related by a power law, then the relationship can be written as:

$$y = k * x^{-\alpha}$$



Graph & Parameter

pdf $\rightarrow$

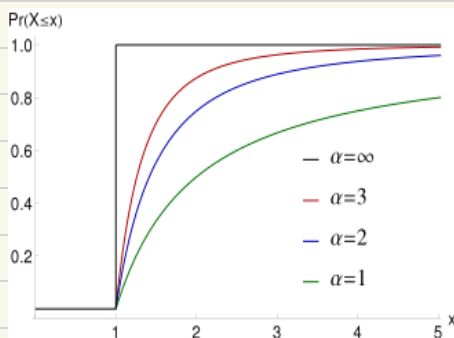


(y is max at x_m)

- eg:
- The size of human settlements
 - File distribution of internet traffic which uses the TCP protocol.

$$\text{PDF} \rightarrow \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$$

Cdf $\rightarrow$



Skewness $\rightarrow$ Skewed.

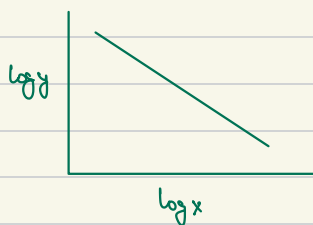
How to detect if a distribution is Pareto Distribution?

- log-log plot

$$y = \frac{a x_m^\alpha}{x^{\alpha+1}}$$

log of x

log of y



← graph.

- Q-Q plot.

$[x, y]$
 Pareto $\hookrightarrow$ compare.

Transformations.

Mathematical Transformation.

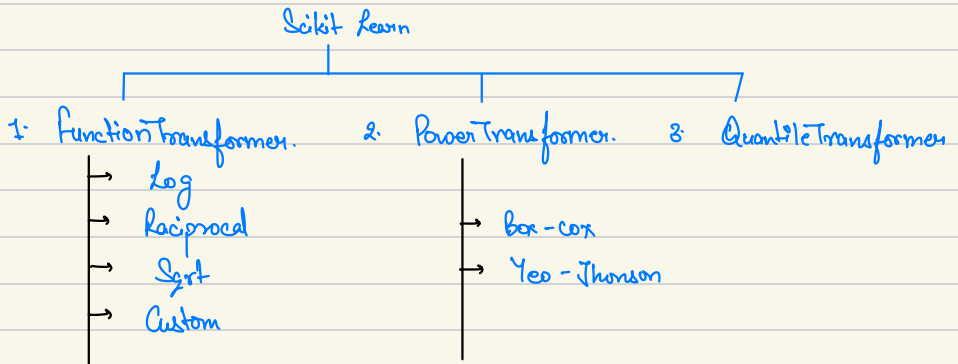
FE

↳ Feature Transformation

↳ Mathematical Transformation.

- Log Transform
- Reciprocal "
- Power (sqrt)

- Box-Cox Transform
- Yeo-Johnson Transform.



How to find if data is normal?

→ Sns.distplot

→ pd.skew() $\Rightarrow 0$ (unskewed)

→ Q-Q plot (most reliable method).

Log Transform.

- Take log of each value
- Generally cannot be applied to -ve value
- Right skewed data is best to apply log-transform.
 - ↳ it brings the data to center. (Q-Q plot).

Reciprocal Transform. ($1/x$)	Sq. Transform (x^2)	Sqrt Transform ($\sqrt{x}$)
• Big values $\rightarrow$ Small values (Viceversa)	• Left Skewed Data.	

Function Transformer Class.

$\rightarrow$ Titanic dataset

- Age
- Fare
- Survived.

`np.log` $\rightarrow$ log cannot work at value 0.

`np.log1p` $\rightarrow$ Adds 1 to data

Power Transform

1. Box-Cox
2. Yeo-Johnson.

1. Box-Cox Transform.

$$x_i(\lambda) = \begin{cases} \frac{x_i^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln(x_i) & \text{if } \lambda = 0 \end{cases}$$

power (pointing to $\lambda \neq 0$)

Internal working of
Box-cox transform ↓

- Maximum likelihood
- Bayesian Stats.

The exponent here is a variable called λ (lambda) that varies over the range of -5 to 5, and in the process of searching, we examine all values of λ . Finally, we choose the optimal value (resulting the best approx. to a normal distribution) for your variable.

❗ Box-cox transformation is strictly applicable to numbers > 0 .

2. Yeo-Johnson Transform.

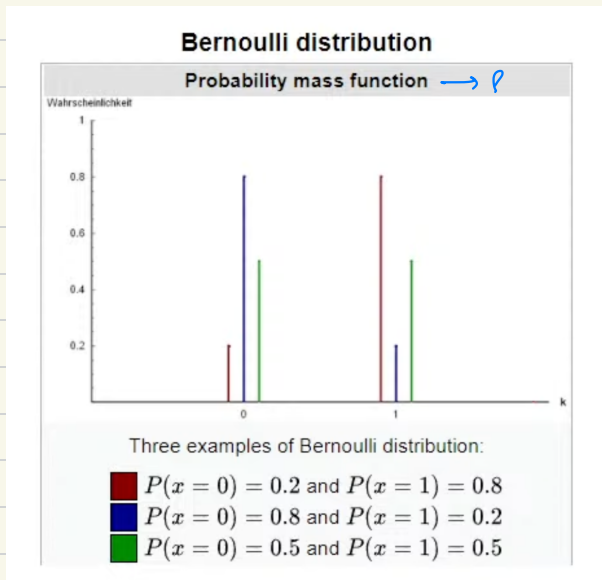
Solves the problem of Box-cox transform (i.e. number > 0)
Use `PowerTransformClass` of `Scikit-learn`.

DISCRETE DISTRIBUTION

Bernoulli Distribution

Bernoulli distribution is a probability distribution that models a binary outcome, where outcome can be either success (represented by 1) or failure (0). The Bernoulli distribution is named after the Swiss mathematician Jacob Bernoulli who introduced it in late 1600s.

It is characterized by a single parameter, which is the probability of success denoted by p .
The PMF of Bernoulli distribution is.



$$P(x) = x = p^x (1-p)^{1-x}$$

p = probability of success
 $1-p$ = prob. of failure.

eg) $P(x=1) = \frac{1}{2}^1 \left(\frac{1}{2}\right)^{1-1}$
Coin toss $\uparrow$
($x=1$)
 $= \frac{1}{2}$

$\downarrow$ ($x=0$)
eg) $P(x=0) = \frac{1}{2}^0 \left(\frac{1}{2}\right)^{1-0}$
 $= \frac{1}{2}$

eg) Coin toss $\rightarrow$ H (1)
 $\rightarrow$ T (0)

eg) Email Classifier $\rightarrow$ Spam (1)
 $\rightarrow$ Not Spam (0)
Determined by

eg) Rolling a dice & getting 5. $\rightarrow 5 \rightarrow \frac{1}{6} \rightarrow 1$
 $\rightarrow \{1, 2, 3, 4, 6\} \rightarrow \frac{5}{6} \rightarrow 0$

Bernoulli distribution is commonly used in ML for modelling binary outcomes.
Mostly for classification of yes or no. eg.) Cat or Dog.

Bernoulli distribution is specifically used in Binomial Distribution.

Binomial Distribution.

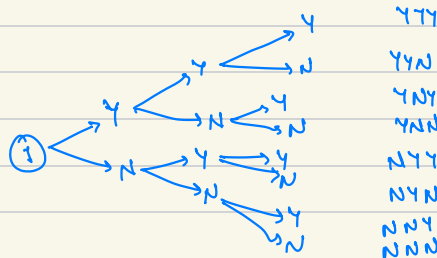
Binomial distribution is a probability distribution that describes the no. of successes in a fixed no. of independent Bernoulli trial with two possible outcomes (commonly called "success" or failure) where the probability of success is constant for each trial. The binomial distribution is characterized by two parameters: the no. of trials (n) and the probability of success (p).

Independent Bernoulli done (n) times becomes Binomial.

Binomial answers the questions like :-

eg.) The probability of watching this lecture & liking it is 0.5. What is prob that :

- (i) No-one out of 3 people will like it. $\rightarrow 1/8$
- (ii) 1 out of 3 people will like it $\rightarrow 3/8$
- (iii) 2 out of 3 people will like it $\rightarrow 3/8$
- (iv) 3 out of 3 people will like it. $\rightarrow 1/8$



$2^3 = 8$ cases

• Not all time it's easy to make sample space, thus we use binomial distribution eqn. :

$$P(X=x) = {}^n C_x p^x (1-p)^{n-x}$$

n = no. of trials.

p = Prob. of success.

x = desired result.

Q1) 2 out of 3 people liking it.

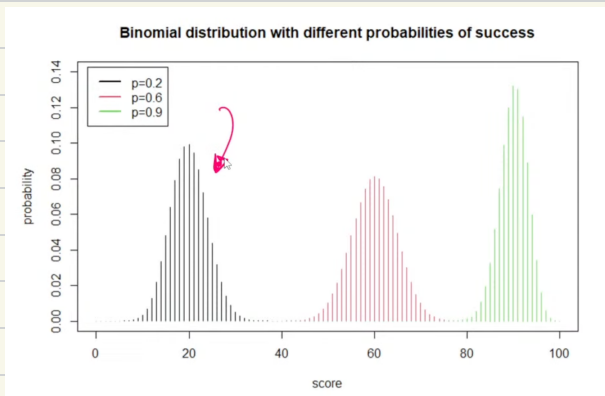
$$= {}^3 C_2 (0.5)^2 (0.5)^{3-2}$$

$$= \frac{3!}{2! 1!} \times \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)$$

$$= \frac{3 \times 2!}{2! \times 1!} \times \frac{1}{8}$$

$$= \frac{3}{8} \text{ Ans.}$$

Pdf →



Criteria : The process consist of n trials.

Only 2 exclusive outcomes are possible, a success & a failure.

$P(\text{success}) = p$ and $P(\text{failure}) = 1-p$ & it is fixed from trial to trial.

The trials are independent.

Uses : Binary Classification Problems.

Hypothesis testing

Logistic regression.

A/B testing.